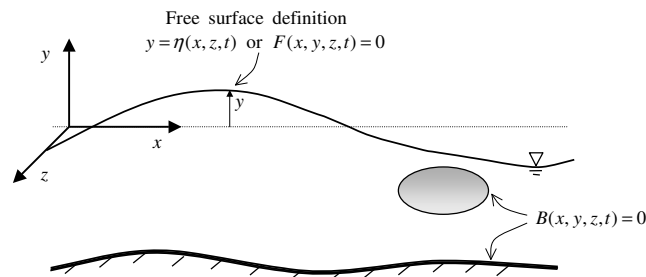


2.20 - Marine Hydrodynamics Lecture 20

Chapter 6 - Water Waves

6.1 Exact (Nonlinear) Governing Equations for Surface Gravity Waves, Assuming Potential Flow



Unknown variables

Velocity field: $\vec{v}(x, y, z, t) = \nabla\phi(x, y, z, t)$

Position of free surface: $y = \eta(x, z, t)$ or $F(x, y, z, t) = 0$

Pressure field: $p(x, y, z, t)$

Governing equations

Continuity: $\nabla^2\phi = 0 \quad y < \eta \text{ or } F < 0$

Bernoulli for P-Flow: $\frac{\partial\phi}{\partial t} + \frac{1}{2}|\nabla\phi|^2 + \frac{p-p_a}{\rho} + gy = 0; \quad y < \eta \text{ or } F < 0$

Far way, no disturbance: $\partial\phi/\partial t, \nabla\phi \rightarrow 0$ and $p = \underbrace{p_a}_{\text{atmospheric}} - \underbrace{\rho gy}_{\text{hydrostatic}}$

Boundary Conditions

1. On an impervious boundary $B(x, y, z, t) = 0$, we have KBC:

$$\vec{v} \cdot \hat{n} = \nabla \phi \cdot \hat{n} = \frac{\partial \phi}{\partial n} = \vec{U}(\vec{x}, t) \cdot \hat{n}(\vec{x}, t) = U_n \text{ on } B = 0$$

Alternatively: a particle P on B remains on B , i.e., B is a material surface. For example if P is on B at $t = t_0$, P stays on B for all t .

$$B(\vec{x}_P, t_0) = 0, \text{ then } B(\vec{x}_P(t), t) = 0 \text{ for all } t,$$

so that, following P B is always 0.

$$\therefore \frac{DB}{Dt} = \frac{\partial B}{\partial t} + (\nabla \phi \cdot \nabla) B = 0 \text{ on } B = 0$$

For example, for a flat bottom at $y = -h \Rightarrow B = y + h = 0 \Rightarrow$

$$\frac{DB}{Dt} = \left(\frac{\partial \phi}{\partial y} \right) \left(\underbrace{\frac{\partial}{\partial y}(y + h)}_{=1} \right) = 0 \Rightarrow \frac{\partial \phi}{\partial y} = 0 \text{ on } B = y + h = 0$$

2. On the free surface, $y = \eta$ or $F = y - \eta(x, z, t) = 0$ we have KBC and DBC.

KBC: free surface is a material surface, no normal velocity relative to the free surface. A particle on the free surface remains on the free surface for all times.

$$\frac{DF}{Dt} = 0 = \frac{D}{Dt}(y - \eta) = \underbrace{\frac{\partial \phi}{\partial y}}_{\substack{\text{vertical} \\ \text{velocity}}} - \frac{\partial \eta}{\partial t} - \frac{\partial \phi}{\partial x} \underbrace{\frac{\partial \eta}{\partial x}}_{\substack{\text{slope} \\ \text{of f.s.}}} - \frac{\partial \phi}{\partial z} \underbrace{\frac{\partial \eta}{\partial z}}_{\substack{\text{slope} \\ \text{of f.s.}}} \text{ on } y = \underbrace{\eta}_{\substack{\text{still} \\ \text{unknown}}}$$

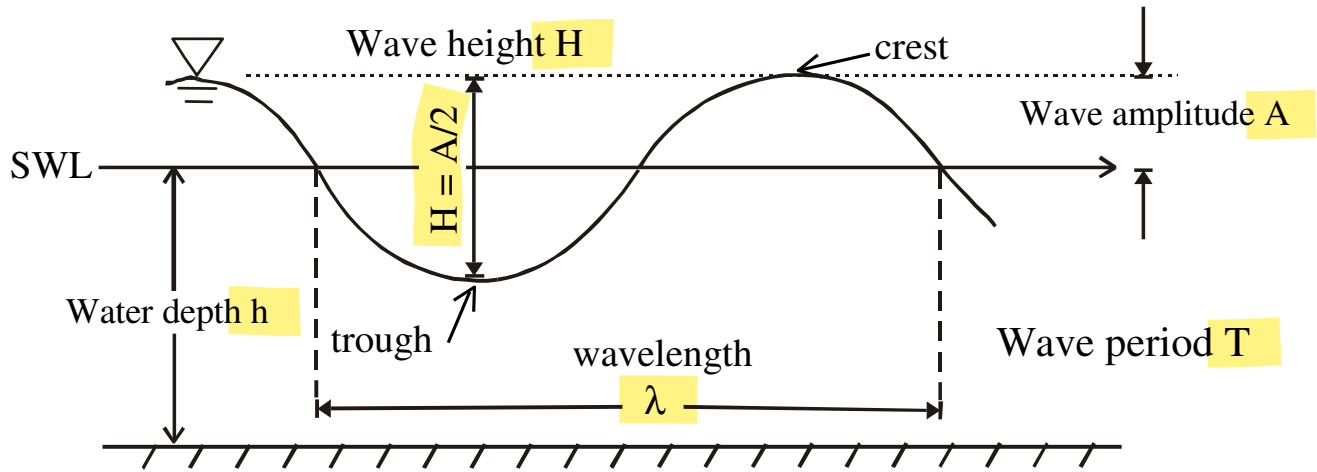
DBC: $p = p_a$ on $y = \eta$ or $F = 0$. Apply Bernoulli equation at $y = \eta$:

$$\frac{\partial \phi}{\partial t} + \underbrace{\frac{1}{2} |\nabla \phi|^2}_{\text{non-linear term}} + g \underbrace{\eta}_{\text{still unknown}} = p_a \text{ on } y = \eta$$

6.2 Linearized (Airy) Wave Theory

Assume small wave amplitude compared to wavelength, i.e., small free surface slope

$$\frac{A}{\lambda} \ll 1$$



Consequently

$$\frac{\phi}{\lambda^2/T}, \frac{\eta}{\lambda} \ll 1$$

We keep only linear terms in ϕ , η .

For example: $(\)|_{y=\eta} = \underbrace{(\)|_{y=0}}_{\text{keep}} + \eta \underbrace{\frac{\partial}{\partial y} (\)|_{y=0}}_{\text{discard}} + \dots$ Taylor series

6.2.1 **BVP** In this paragraph we state the Boundary Value Problem for linear (Airy) waves.

Finite depth $h = \text{const}$	Infinite depth
GE: $\nabla^2 \phi = 0, \quad -h < y < 0$	$\nabla^2 \phi = 0, \quad y < 0$
BKBC: $\frac{\partial \phi}{\partial y} = 0, \quad y = -h$	$\nabla \phi \rightarrow 0, \quad y \rightarrow -\infty$
FSKBC: $\frac{\partial \phi}{\partial y} = \frac{\partial \eta}{\partial t}, \quad y = 0$ FSDBK: $\frac{\partial \phi}{\partial t} + g\eta = 0, \quad y = 0$	$\left. \begin{array}{l} \frac{\partial \phi}{\partial y} = \frac{\partial \eta}{\partial t}, \quad y = 0 \\ \frac{\partial \phi}{\partial t} + g\eta = 0, \quad y = 0 \end{array} \right\} \rightarrow \frac{\partial^2 \phi}{\partial t^2} + g \frac{\partial \phi}{\partial y} = 0$

Introducing the notation $\{\}$ for infinite depth we can rewrite the BVP:

$$\begin{array}{ll} \text{Constant finite depth } h & \{\text{Infinite depth}\} \\ \nabla^2 \phi = 0, \quad -h < y < 0 & \{\nabla^2 \phi = 0, \quad y < 0\} \end{array} \quad (1)$$

$$\begin{array}{ll} \frac{\partial \phi}{\partial y} = 0, \quad y = -h & \{\nabla \phi \rightarrow 0, \quad y \rightarrow -\infty\} \end{array} \quad (2)$$

$$\begin{array}{ll} \frac{\partial^2 \phi}{\partial t^2} + g \frac{\partial \phi}{\partial y} = 0, \quad y = 0 & \left\{ \frac{\partial^2 \phi}{\partial t^2} + g \frac{\partial \phi}{\partial y} = 0, \quad y = 0 \right\} \end{array} \quad (3)$$

Given ϕ calculate:

$$\eta(x, t) = -\frac{1}{g} \frac{\partial \phi}{\partial t} \Big|_{y=0} \quad \left\{ \eta(x, t) = -\frac{1}{g} \frac{\partial \phi}{\partial t} \Big|_{y=0} \right\} \quad (4)$$

$$p - p_a = \underbrace{-\rho \frac{\partial \phi}{\partial t}}_{\text{dynamic}} - \underbrace{\rho g y}_{\text{hydrostatic}} \quad \left\{ p - p_a = \underbrace{-\rho \frac{\partial \phi}{\partial t}}_{\text{dynamic}} - \underbrace{\rho g y}_{\text{hydrostatic}} \right\} \quad (5)$$

6.2.2 **Solution** Solution of 2D periodic plane progressive waves, applying separation of variables.

We seek solutions to Equation (1) of the form $e^{i\omega t}$ with respect to time. Using the KBC (2), after some algebra we find ϕ . Upon substitution in Equation (4) we can also obtain η .

$$\phi = \frac{gA}{\omega} \sin(kx - \omega t) \frac{\cosh k(y+h)}{\cosh kh}$$

$$\left\{ \phi = \frac{gA}{\omega} \sin(kx - \omega t) e^{ky} \right\}$$

$$\eta \underset{\text{using (4)}}{=} A \cos(kx - \omega t)$$

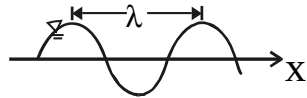
$$\left\{ \eta \underset{\text{using (4)}}{=} A \cos(kx - \omega t) \right\}$$

where A is the wave amplitude $A = H/2$.

Exercise Verify that the obtained values for ϕ and η satisfy Equations (1), (2), and (4).

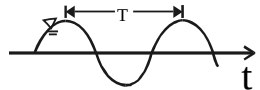
6.2.3 **Review on plane progressive waves**

- (a) At $t = 0$ (say), $\eta = A \cos kx \rightarrow$ periodic in x with **wavelength: $\lambda = 2\pi/k$**
Units of λ : $[L]$



$$k = \text{wavenumber} = 2\pi/\lambda \quad [L^{-1}]$$

- (b) At $x = 0$ (say), $\eta = A \cos \omega t \rightarrow$ periodic in t with **period: $T = 2\pi/\omega$**
Units of T : $[T]$



$$\omega = \text{frequency} = 2\pi/T \quad [T^{-1}], \text{ e.g. rad/sec}$$

- (c) $\eta = A \cos \left[k \left(x - \frac{\omega}{k} t \right) \right]$
Units of $\frac{\omega}{k}$: $\left[\frac{L}{T} \right]$

Following a point with velocity $\frac{\omega}{k}$, i.e., $x_p = \left(\frac{\omega}{k} \right) t + \text{const}$, the phase of η does not change, i.e., $\frac{\omega}{k} = \frac{\lambda}{T} \equiv V_p \equiv \text{phase velocity}$.

6.2.4 Dispersion Relation

So far, any ω, k combination is allowed. However, recall that we still have not made use of the FSBC Equation (3). Upon substitution of ϕ in Equation (3) we find that the following relation between h, k , and ω must hold:

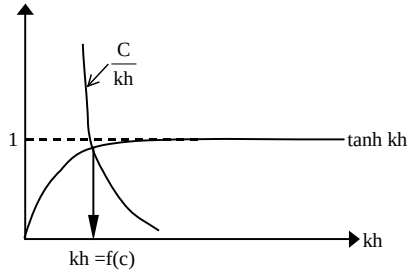
$$\frac{\partial^2 \phi}{\partial t^2} + g \frac{\partial \phi}{\partial y} = 0 \quad \xrightarrow[\phi = \frac{gA}{\omega} \sin(kx - \omega t) f(z)]{\quad} \quad -\omega^2 \cosh kh + gk \sinh kh = 0 \Rightarrow \omega^2 = gk \tanh kh$$

- This is the **Dispersion Relation**

$$\omega^2 = gk \tanh kh \quad \quad \quad \{\omega^2 = gk\} \quad (6)$$

Given h , the Dispersion Relation (6) provides a **unique** relation between ω and k , i.e., $\omega = \omega(k; h)$ or $k = k(\omega; h)$.

- Proof



$$C \equiv \frac{\omega^2 h}{g} \underbrace{\quad}_{\text{from (6)}} (kh) \tanh (kh)$$

$$\frac{C}{kh} = \tanh kh$$

→ obtain unique solution for k

- Comments

- *General* As $\omega \uparrow$ then $k \uparrow$, or equivalently as $T \uparrow$ then $\lambda \uparrow$.

- *Phase speed* $V_p \equiv \frac{\lambda}{T} = \frac{\omega}{k} = \sqrt{\frac{g}{k} \tanh kh} \quad \left\{ V_p = \sqrt{\frac{g}{k}} \right\}$

Therefore as $T \uparrow$ or as $\lambda \uparrow$, then $V_p \uparrow$, i.e., longer waves are ‘faster’ in terms of phase speed.

- *Water depth effect* For waves the same k (or λ), at different water depths, as $h \uparrow$ then $V_p \uparrow$, i.e., for fixed k V_p is fastest in deep water.
- *Frequency dispersion* Observe that $V_p = V_p(k)$ or $V_p(\omega)$. This means that waves of different frequencies, have different phase speeds, i.e., frequency dispersion.

6.2.5 Solutions to the Dispersion Relation : $\omega^2 = gk \tanh kh$

Property of $\tanh kh$:

$$\tanh kh = \frac{\sinh kh}{\cosh kh} = \frac{1 - e^{-2kh}}{1 + e^{-2kh}} \cong \begin{cases} kh & \text{for } kh \ll 1. \text{ In practice } \overbrace{h < \lambda/20}^{\text{long waves shallow water}} \\ 1 & \text{for } kh \gtrsim 3. \text{ In practice } \underbrace{h > \frac{\lambda}{2}}_{\text{short waves deep water}} \end{cases}$$

Shallow water waves or long waves	Intermediate depth or wavelength	Deep water waves or short waves
$kh \ll 1$ $\sim h < \lambda/20$	Need to solve $\omega^2 = gk \tanh kh$ given ω, h for k (given k, h for ω - easy!)	$kh \gg 1$ $\sim h > \lambda/2$
$\omega^2 \cong gk \cdot kh \rightarrow \omega = \sqrt{gh} k$ $\lambda = \sqrt{gh} T$	(a) Use tables or graphs (e.g. JNN fig.6.3) $\omega^2 = gk \tanh kh = gk_\infty$ $\Rightarrow \frac{k_\infty}{k} = \frac{\lambda}{\lambda_\infty} = \frac{V_p}{V_{p\infty}} = \tanh kh$ (b) Use numerical approximation (hand calculator, about 4 decimals) i. Calculate $C = \omega^2 h / g$ ii. If $C > 2$: "deeper" $\Rightarrow$ $kh \approx C(1 + 2e^{-2C} - 12e^{-4C} + \dots)$ If $C < 2$: "shallower" $\Rightarrow$ $kh \approx \sqrt{C}(1 + 0.169C + 0.031C^2 + \dots)$	$\omega^2 = gk$ $\lambda = \frac{g}{2\pi} T^2$ $(\lambda(\text{in ft.}) \approx 5.12 T^2 \text{ (in sec.)})$
No frequency dispersion $V_p = \sqrt{gh}$	Frequency dispersion $V_p = \sqrt{\frac{g}{k} \tanh kh}$	Frequency dispersion $V_p = \sqrt{\frac{g}{2\pi} \lambda}$

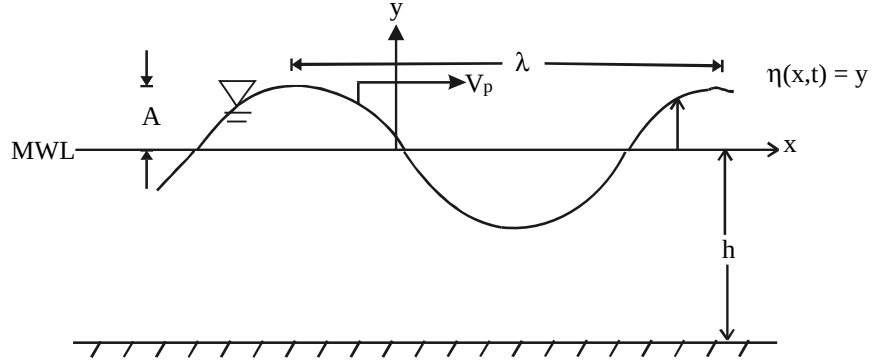
6.3 Characteristics of a Linear Plane Progressive Wave

$$k = \frac{2\pi}{\lambda}$$

$$\omega = \frac{2\pi}{T}$$

$$H = 2A$$

Define $U \equiv \omega A$

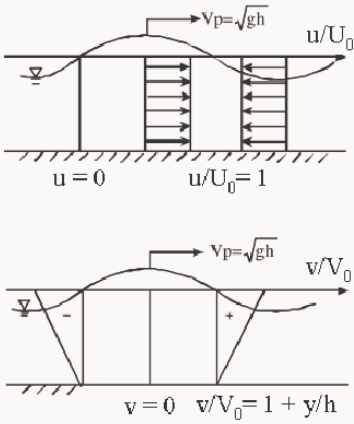
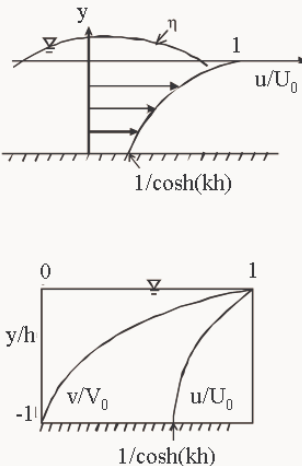
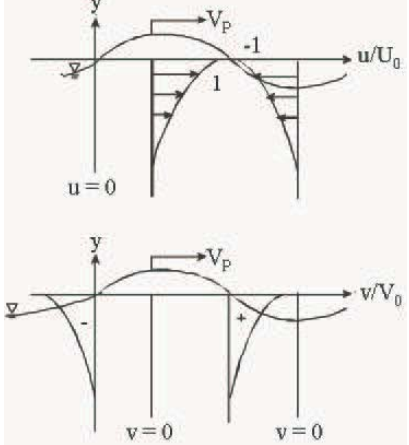


Linear Solution:

$$\eta = A \cos(kx - \omega t); \quad \phi = \frac{Ag}{\omega} \frac{\cosh k(y+h)}{\cosh kh} \sin(kx - \omega t), \text{ where } \omega^2 = gk \tanh kh$$

6.3.1 Velocity field

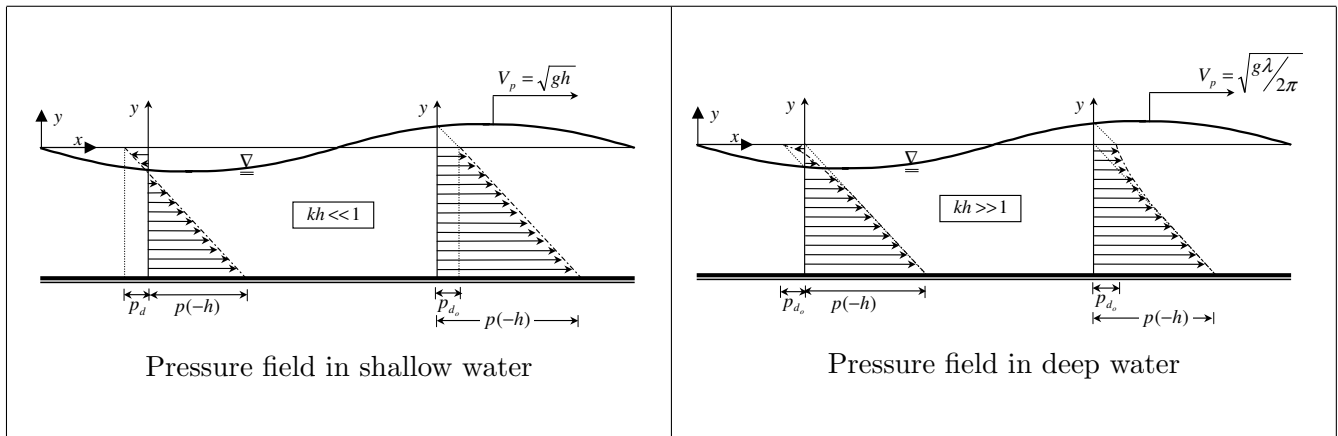
Velocity on free surface $\vec{v}(x, y = 0, t)$	
$u(x, 0, t) \equiv U_o = A\omega \frac{1}{\tanh kh} \cos(kx - \omega t)$	$v(x, 0, t) \equiv V_o = A\omega \sin(kx - \omega t) = \frac{\partial \eta}{\partial t}$
Velocity field $\vec{v}(x, y, t)$	
$u = \frac{\partial \phi}{\partial x} = \frac{Agk}{\omega} \frac{\cosh k(y+h)}{\cosh kh} \cos(kx - \omega t)$ $= \underbrace{A\omega}_U \frac{\cosh k(y+h)}{\sinh kh} \cos(kx - \omega t) \Rightarrow$ $\frac{u}{U_o} = \frac{\cosh k(y+h)}{\cosh kh} \begin{cases} \sim e^{ky} & \text{deep water} \\ \sim 1 & \text{shallow water} \end{cases}$	$v = \frac{\partial \phi}{\partial y} = \frac{Agk}{\omega} \frac{\sinh k(y+h)}{\cosh kh} \sin(kx - \omega t)$ $= \underbrace{A\omega}_U \frac{\sinh k(y+h)}{\sinh kh} \sin(kx - \omega t) \Rightarrow$ $\frac{v}{V_o} = \frac{\sinh k(y+h)}{\sinh kh} \begin{cases} \sim e^{ky} & \text{deep water} \\ \sim 1 + \frac{y}{h} & \text{shallow water} \end{cases}$
• u is in phase with η	• v is out of phase with η

Velocity field $\vec{v}(x, y)$		
Shallow water	Intermediate water	Deep water
 $u = 0$ $u/U_0 = 1$ $v = 0$ $v/V_0 = 1 + y/h$	 $1/\cosh(kh)$ $1/\cosh(kh)$ Shallow water / Long waves: $kh \ll 1$ $u = \frac{A\omega}{kh} \cos(kx - \omega t) = \eta \sqrt{\frac{g}{h}}$ $v = A\omega \left(1 + \frac{y}{h}\right) \sin(kx - \omega t)$	 $u = 0$ $u/U_0 = 1$ $v = 0$ $v/V_0 = 1$ Rule of thumb $\frac{u}{u_s} = \frac{v}{v_s} \approx 4\%$ at $y = -\frac{\lambda}{2}$ $(\cosh kh \sim 1, \sinh kh \sim kh)$

6.3.2 Pressure field

- Total pressure $p = p_d - \rho g y$.
- Dynamic pressure $p_d = -\rho \frac{\partial \phi}{\partial t}$.
- Dynamic pressure on free surface $p_d(x, y = 0, t) \equiv p_{d_o}$

Pressure field		
Shallow water	Intermediate water	Deep water
$p_d = \rho g \eta$	$p_d = \rho g A \frac{\cosh k(y+h)}{\cosh kh} \cos(kx - \omega t)$ $= \rho g \frac{\cosh k(y+h)}{\cosh kh} \eta$	$p_d = \rho g e^{ky} \eta$
$\frac{p_d}{p_{d_o}}$ same picture as $\frac{u}{U_o}$		
$\frac{p_d(-h)}{p_{d_o}} = 1$ (no decay)	$\frac{p_d(-h)}{p_{d_o}} = \frac{1}{\cosh kh}$	$\frac{p_d(-h)}{p_{d_o}} = e^{-ky}$
$p = \underbrace{\rho g(\eta - y)}_{\text{"hydrostatic" approximation}}$		$p = \rho g (\eta e^{ky} - y)$



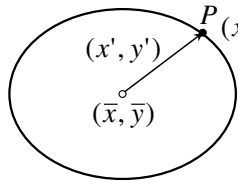
6.3.3 Particle Orbits (‘Lagrangian’ concept)

Let $x_p(t), y_p(t)$ denote the position of particle P at time t .

Let $(\bar{x}; \bar{y})$ denote the mean position of particle P.

The position P can be rewritten as $x_p(t) = \bar{x} + x'(t)$, $y_p(t) = \bar{y} + y'(t)$, where $(x'(t), y'(t))$ denotes the departure of P from the mean position.

In the same manner let $\vec{v} \equiv \vec{v}(\bar{x}, \bar{y}, t)$ denote the velocity at the mean position and $\vec{v}_p \equiv \vec{v}(x_p, y_p, t)$ denote the velocity at P.



$$\begin{aligned} \vec{v}_p &= \vec{v}(\bar{x} + x', \bar{y} + y', t) \xrightarrow{\text{TSE}} \\ \vec{v}_p &= \vec{v}(\bar{x}, \bar{y}, t) + \underbrace{\frac{\partial \vec{v}}{\partial x}(\bar{x}, \bar{y}, t) x' + \frac{\partial \vec{v}}{\partial y}(\bar{x}, \bar{y}, t) y' + \dots}_{\text{ignore - linear theory}} \\ \vec{v}_p &\cong \vec{v} \end{aligned}$$

To estimate the position of P, we need to evaluate $(x'(t), y'(t))$:

$$\begin{aligned} x' &= \int dt u(\bar{x}, \bar{y}, t) = \int dt \omega A \frac{\cosh k(\bar{y} + h)}{\sinh kh} \cos(k\bar{x} - \omega t) \Rightarrow \\ &= -A \frac{\cosh k(\bar{y} + h)}{\sinh kh} \sin(k\bar{x} - \omega t) \\ y' &= \int dt v(\bar{x}, \bar{y}, t) = \int dt \omega A \frac{\sinh k(\bar{y} + h)}{\sinh kh} \sin(k\bar{x} - \omega t) \Rightarrow \\ &= A \frac{\sinh k(\bar{y} + h)}{\sinh kh} \cos(k\bar{x} - \omega t) \end{aligned}$$

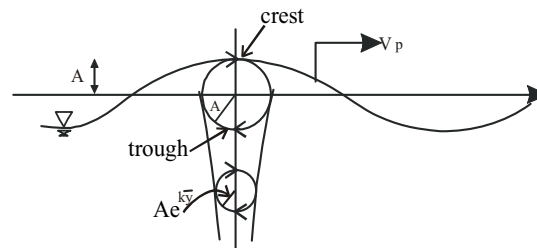
Check: On $\bar{y} = 0$, $y' = A \cos(k\bar{x} - \omega t) = \eta$, i.e., the vertical motion of a free surface particle (in linear theory) coincides with the vertical free surface motion.

It can be shown that the particle motion satisfies

$$\frac{x'^2}{a^2} + \frac{y'^2}{b^2} = 1 \Leftrightarrow \frac{(x_p - \bar{x})^2}{a^2} + \frac{(y_p - \bar{y})^2}{b^2} = 1$$

where $a = A \frac{\cosh k(\bar{y} + h)}{\sinh kh}$ and $b = A \frac{\sinh k(\bar{y} + h)}{\sinh kh}$, i.e., the particle orbits form closed ellipses with horizontal and vertical axes a and b .

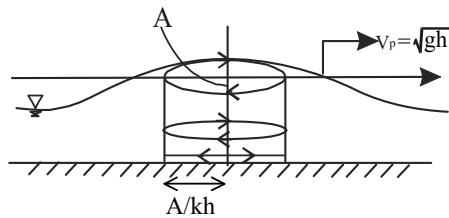
(a) deep water $kh \gg 1$: $a = b = Ae^{ky}$
 circular orbits with radii Ae^{ky} decreasing
 exponentially with depth



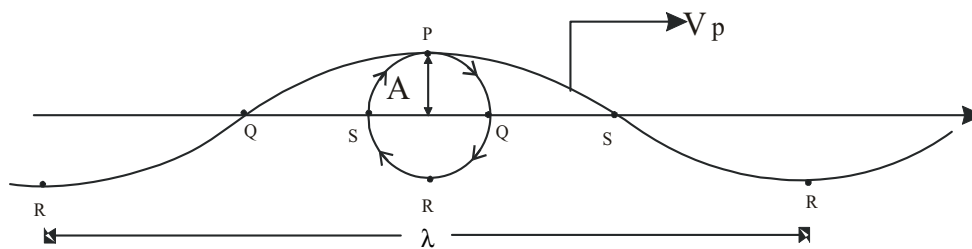
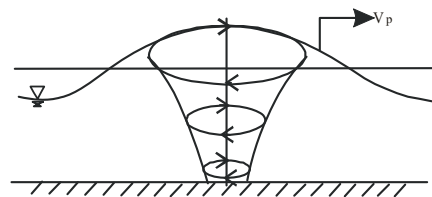
(b) shallow water $kh \ll 1$:

$$a = \frac{A}{kh} = \text{const.}; b = A\left(1 + \frac{y}{h}\right)$$

decreases linearly
 with depth



(c) Intermediate depth



6.3.4 Summary of Plane Progressive Wave Characteristics

$f(y)$	Deep water/ short waves $kh > \pi$ (say)	Shallow water/ long waves $kh \ll 1$
$\frac{\cosh k(y+h)}{\cosh kh} = f_1(y) \sim$ e.g. p_d	e^{ky}	1
$\frac{\cosh k(y+h)}{\sinh kh} = f_2(y) \sim$ e.g. u, a	e^{ky}	$\frac{1}{kh}$
$\frac{\sinh k(y+h)}{\sinh kh} = f_3(y) \sim$ e.g. v, b	e^{ky}	$1 + \frac{y}{h}$

$C(x) = \cos(kx - \omega t)$ (in phase with η)	$S(x) = \sin(kx - \omega t)$ (out of phase with η)
$\frac{\eta}{A} = C(x)$	
$\frac{u}{A\omega} = C(x) f_2(y)$	$\frac{v}{A\omega} = S(x) f_3(y)$
$\frac{p_d}{\rho g A} = C(x) f_1(y)$	
$\frac{y'}{A} = C(x) f_3(y)$	$\frac{x'}{A} = -S(x) f_2(y)$
$\frac{a}{A} = f_2(y)$	$\frac{b}{A} = f_3(y)$

